



Projectiles · Mark Scheme

Mechanics Chapter 7: Projectiles. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Horizontal projection

- 1 $0.8 = 4.9t^2$ M1; $t = 0.404$ s A1.
- 2 $20 = 4.9t^2$, $t = 2.02$ s M1 A1; $x = 12 \times 2.02 = 24.2$ m A1.
- 3 $45 = 4.9t^2$, $t = 3.03$ M1 A1; $U = \frac{30}{3.03} = 9.90 \text{ ms}^{-1}$ A1.
- 4 $t = \sqrt{\frac{80}{9.8}} = 2.857$, $v_y = 28$ M1 A1; speed $\sqrt{10^2 + 28^2} = 29.7 \text{ ms}^{-1}$ A1; angle $\tan^{-1} \frac{28}{10} = 70.3^\circ$ below the horizontal A1.
- 5(a) $t = \frac{2.4}{18} = 0.133$ s B1.
- 5(b) $s = 4.9 \times 0.133^2$ M1; $= 0.0871$ m (8.71 cm) A1.
- 5(c) $0.05 = 4.9t^2$ M1; $t = 0.101$ s A1; $U = \frac{2.4}{0.101} = 23.8 \text{ ms}^{-1}$ A1.
- 5(d) E.g. no air resistance / the dart is a particle B1.

Section B – At an angle

- 6 $20 \cos 30^\circ = 17.3 \text{ ms}^{-1}$ B1; $20 \sin 30^\circ = 10 \text{ ms}^{-1}$ B1.
- 7 $0 = 10t - 4.9t^2$ M1; $t = 2.04$ s A1; range $17.32 \times 2.04 = 35.3$ m A1.
- 8 $0 = (25 \sin 50^\circ)^2 - 2(9.8)h$ M1 A1; $h = 18.7$ m A1.
- 9 $T = \frac{2 \times 30 \sin 40^\circ}{9.8}$ M1; $= 3.94$ s A1; $R = 30 \cos 40^\circ \times 3.935$ M1; $= 90.4$ m A1.
- 10(a) Horizontal speed $24 \times 0.8 = 19.2$ B1; $t = \frac{30}{19.2} = 1.5625$ s B1.
- 10(b) Vertical speed $24 \times 0.6 = 14.4$ B1; $y = 14.4(1.5625) - 4.9(1.5625)^2$ M1; $= 10.5$ m A1.
- 10(c) $v_y = 14.4 - 9.8(1.5625)$ M1; $= -0.9125$, so falling A1.
- 10(d) $\sqrt{19.2^2 + 0.9125^2}$ M1; $= 19.2 \text{ ms}^{-1}$ A1.

Section C – Trajectory equation

- 11 $x = 10 \cos 45^\circ t$, $y = 10 \sin 45^\circ t - 4.9t^2$ M1; eliminate t : $y = x - \frac{4.9x^2}{50} = x - 0.098x^2$ A1.
- 12 $x(1 - 0.098x) = 0$ M1; range 10.2 m A1; $y(5) = 5 - 2.45 = 2.55$ m A1.
- 13 $x = U \cos \theta t \Rightarrow t = \frac{x}{U \cos \theta}$ M1; $y = U \sin \theta t - \frac{1}{2}gt^2$ M1; substitute and simplify to the given result A1.
- 14 Using the trajectory equation with $x = 12$ M1; $y = 12 \tan 35^\circ - \frac{9.8 \times 144}{2 \times 225 \cos^2 35^\circ} = 8.40 - 4.67$ M1; $= 3.73$ m above the point of release, i.e. 5.23 m above the ground A1; clears the fence by 2.23 m A1.
- 15(a) Taking up as positive: $-25 = 10t - 4.9t^2$ M1 A1; $t = 3.50$ s A1.
- 15(b) $x = 20 \cos 30^\circ \times 3.499$ M1; $= 60.6$ m A1.
- 15(c) $v_x = 17.3$, $v_y = 10 - 9.8(3.499) = -24.3$ M1 A1; speed 29.8 ms^{-1} A1; 54.5° below the horizontal A1.

Roar-level challenge



16 $R = \frac{U^2 \sin 2\theta}{g}$ *M1*; $\sin 2\theta = \frac{40 \times 9.8}{441} = 0.889$ *A1*; $2\theta = 62.7^\circ$ or 117.3° *M1*; $\theta = 31.4^\circ$ or 58.6° *A1*; 31.4° gives the shorter time, since $T = \frac{2U \sin \theta}{g}$ is smaller *B1*.